

1. Integrar:

a) $\int \frac{3 dx}{17 \cosh x - 8 \sinh x + 15}$

(2p)

b) $\int \frac{(2 + 3 \cos x) dx}{\cos x (1 + 4 \cos x)}$

(2p)

2. Integrar:

a) $\int \frac{\sin x \cdot \sin 2x}{\sin^4 x + \cos^4 x + 1} dx$

(3p)

b) $\int \frac{dx}{(x+2)^2 \sqrt{x^2 + 2x - 1}}$

(3p)

3. Sea $I(m, n) = \int \frac{\sin^{n+1} x}{\cos^{m+1} x} dx$; $m, n \in \mathbb{Z}^+$

· Demostrar que:

$$I(m, n) = \frac{\sin^n x}{m \cos^m x} - \frac{n}{m} I(m-2, n-2) \quad ; \quad m, n \in \mathbb{Z} ; m, n \geq 3 \quad (3p)$$

4. Integrar:

a) $\int \frac{dx}{\sqrt{x} (\sqrt{x} + 1)^{2/3}}$

(2p)

b) $\int x \cdot \sqrt[3]{1 + x^{-2}} dx$

(2p)

5. Integrar:

$$\int \frac{\sqrt{3 - 12x - 6x^2}}{2x^5 + 4x^4 - x^3} dx$$

(3p)